Shengqing Xia *Phone:* **(765)337-4158** *E-mail:* **xia170@purdue.edu**

Education

Purdue University, West Lafayette, IN, USA **Aug. 2021 – Now**
Ph.D., Computer Science, GPA: 4.0/4.0
Advisor: Chunyi Peng

Columbia University, New York, NY, USA **Aug. 2019 – Feb. 2021**
M.S., Data Science, GPA: 3.943/4.0

Columbia University, New York, NY, USA **Jan. 2018 – May. 2018**
Exchange Student, Statistics, GPA: 3.917/4.0

Wuhan University, Wuhan, China **Sep. 2015 – Jul. 2019**
B.E., Printing Engineering, GPA: 3.87/4.0 (Top 1)

Professional Experience

Alibaba Cloud, Hangzhou, China **Mar. 2021 – Jul. 2021**
Algorithm Engineer (Full-time)
- Designed anomaly detection algorithms for 130 min KPIs data based on LSTM model.

Tencent (CSIG), Shenzhen, China **Jun. 2019 – Jul. 2019**
Big Data and Algorithm Application Intern
- Created an automatic grading system for high school compositions with Transformer.

Awards

- 2026 MobiCom 2026 N2Women Student Travel Grant
- 2026 IROS 2026 IEEE RAS Travel Grant
- 2026 Purdue Graduate Teaching Award
- 2025 Purdue Teaching Fellowship
- 2024 ACM HotMobile Best Demo Award
- 2024 ACM HotMobile Student Travel Grant

Publications

[1] **S. Xia** and C. Peng. *"Depth4vtol: Precise Depth Estimation for Vertical Take-off and Landing UAVs over Residential Houses"*, in ACM ImmerCom 2026 (Workshop co-located with ACM MobiCom 2026),

[2] **S. Xia**, J. Du, and C. Peng. *"FollowMe: Automating Precise Image Annotation Across Various Environments"*, in ACM HumanSys 2026 (Workshop co-located with ACM MobiCom 2026),

[3] **S. Xia**, J. Du, and C. Peng. *"AnchorDepth: Fast Metric Depth Estimation with Anchors for Drones"*, in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026, video link: https://drive.google.com/file/d/1j01guvY2mkAPHhFg0A-hA7k0-Ye4rAu5,

[4] J. Du, **S. Xia**, and C. Peng. *"A4AWD: Augmenting Aerial Weed Detection over Residential Lawns"*, in IEEE International Conference on Unmanned Aircraft Systems (ICUAS) 2026,

[5] **S. Xia**, J. Du, and C. Peng. *"RSSS: Robust Structural Semantic Segmentation for Autonomous Drone Delivery to Door"*, in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025,

[6] **S. Xia**, J. Guo, and C. Peng. *"SSS: Towards Autonomous Drone Delivery to Your Door Over House-Aware Semantics"*, in ACM HotMobile 2024,

[7] **S. Xia**, J. Guo, and C. Peng. *"Demo: Towards Autonomous Drone Delivery to Your Door Over House-Aware Semantics"*, in ACM HotMobile Demos 2024, demo link: https://ddd.cs.purdue.edu, video link: https://github.com/mssn/dd-demo/blob/main/demo.mp4,

[8] J. Guo, **S. Xia (co-primary author)**, and C. Peng. *"OPA: One-Predict-All For Efficient Deployment"*, in IEEE INFOCOM 2023,

[9] J. Guo, **S. Xia**, and C. Peng. *"VPPlus: Exploring the Potentials of Video Processing for Live Video Analytics at the Edge"*, in IEEE IWQoS 2022

[10] Z. Zhu, X. Li, Z. Wang, L. He, B. He, **S. Xia**. *"Development and Research of a Multi-Medium Motion Capture System for Underwater Intelligent Agents"*, Appl. Sci. 2020, 10, 6237.

[11] F. Li, B. He, G. Li, M. Ma, J. Li and **S. Xia**. *"Real-time CAD-level surface of revolution reconstruction on visual slam"*, in CAC 2020.

Research Experience

Purdue University, Research Assistant West Lafayette, IN, USA
Advisor: Prof.Chunyi Peng

- Developed a precise metric depth estimation framework for safe drone navigation [3] and landing [1] using a single RGB camera.
- Developed robust semantic segmentation [4] and automated image annotation techniques [2] to enable reliable autonomous drone delivery under diverse environmental conditions
- Built a prototype for autonomous drone delivery [5],[6].
- Designed a deep learning system with one trained super-network and select its sub-network to fit the target deployment scenario for computer vision applications [7].
- Created a system to fulfill the potential to compress video while retaining analytical accuracy [8].

Columbia University, Research Assistant New York, NY, USA
Advisor: Prof. Chong Li

- Built a platform to use automatic text segmentation and summarization to extract outlines.

Tongji University, Research Assistant Shanghai, China
Advisor: Prof. Bin He

- Designed a system to do 3D reconstruction [7] and movement tracing for underwater agents [6].

Columbia University, Team member New York, NY, USA
Advisor: Prof. Pierre Elias and Prof. Adler Perotte

- Analyzed electrocardiogram using deep learning models for the diagnosis of cardiac amyloidosis.